# José Carlos Ruiz Sánchez

## Full-Stack Developer

Node.js | Express.js | React | Next.js | TypeScript | PostgreSQL | PostGIS | Docker

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### SUMMARY

Full-stack developer with a BSc in Computer Engineering and an MSc in Computer Engineering (Big Data) in progress at the University of Almería. During a six-month internship at the Applied Computing Group (ACG), I led most of the development of UAL Trace, a web app that maps campus occupancy from Wi-Fi-connected devices with data refreshed every two minutes. I worked across the PostGIS data model, the n8n ingestion pipeline, the Node.js/Express REST API, the Next.js front end and the Docker deployment. On my own initiative I built Tokimori, a full-stack app with six Node.js/TypeScript microservices behind an nginx API gateway and a React 19 front end.

### TECHNICAL SKILLS

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|---|---|
| **Languages** | JavaScript, TypeScript, Python, SQL |
| **Back end** | Node.js, Express.js, Flask, REST APIs, OpenAPI/Swagger, microservices |
| **Front end** | React, Next.js, Vite, React Router, Leaflet, D3.js, Chart.js, HTML5 Canvas, CSS Modules, Tailwind CSS |
| **Databases** | PostgreSQL, PostGIS, MySQL |
| **DevOps** | Docker, Docker Compose, nginx (reverse proxy, API gateway), Git |
| **Testing** | Vitest, Jest, Supertest, React Testing Library |
| **Security** | JWT, bcrypt, helmet, rate limiting, CORS |
| **Data and automation** | n8n, Spark |
| **Academic coursework** | Java, Spring, JUnit, Angular, R, MongoDB, Cassandra, Power BI |

### EXPERIENCE

**Full-Stack Developer (Extracurricular Internship)** Mar 2026 - Sep 2026

Applied Computing Group (ACG), University of Almería | Almería, Spain | 25 h/week, on-site and remote

- Took over a Next.js prototype running on mock data and led most of the remaining development of UAL Trace, coordinating daily with the group lead through feature branches and code reviews.
- Designed the PostgreSQL/PostGIS data model (12 tables, spatial and time-window indexes, versioned migrations) and an n8n pipeline ingesting ~3,000 rows every 2 minutes from the Cisco Meraki location API, with device identifiers pseudonymised using HMAC-SHA512.
- Built a REST API in Node.js, Express.js and TypeScript (~30 operations, documented with OpenAPI/Swagger) covered by more than 200 tests with Vitest and Supertest.
- Wrote about 80% of the current Next.js front end, including Leaflet and D3.js map layers (hexagon grid, heatmap, building occupancy, routes between buildings, floor-level detail), a historical mode with time playback, statistics and PDF reports, covered by component tests with React Testing Library.
- Optimised performance under real load with an in-memory cache that merges identical concurrent requests, the Node.js cluster module, covering indexes and a single endpoint that replaced up to 13 sequential requests.
- Deployed with Docker multi-stage builds behind an nginx reverse proxy.

## PROJECTS

### Tokimori
Feb 2026 - Aug 2026

Personal project | Full-stack web application | Demo video on YouTube, source code available on request

- Six microservices (auth, items, collection, notes, objectives/canvas, sessions) in Node.js, Express 5 and TypeScript over a shared 8-table MySQL database, behind an nginx API gateway as the only public entry point.
- React 19 front end with Vite, React Router and TypeScript, including a visual editor built on the HTML5 Canvas API with no external libraries that supports shapes, images, text, free drawing, rotation, layers and undo/redo.
- JWT authentication over secure cookies with bcrypt, helmet, rate limiting and configurable CORS.
- Tested with Vitest and Testing Library on the front end and Jest and Supertest on the back end.
- Containerised with Docker Compose, with separate production and development configurations.

### Auscultify
Dec 2024 - Jan 2026

Bachelor's thesis | Graded 9.4/10 | Defended Feb 2026 | github.com/jrs407/Auscultify

- Web app for medical students to practise auscultation with real heart, lung and abdominal recordings in 10-question sessions, with personal statistics, daily streaks and comparison with followed users.
- Nine microservices, eight in Node.js, Express and TypeScript and one in Python and Flask for the recommendation engine, with MySQL 8 and Docker Compose.
- Recommendation system with 11 algorithms, from cold-start baselines to user-based and item-based collaborative filtering with Pearson correlation.
- Full analysis and design (27 functional requirements, use cases, Figma prototypes), a React 19 and TypeScript front end with Chart.js, and two technical reports written in LaTeX.

### PuppyClicker
Apr 2025 - Jun 2025

Personal project | Full-stack browser game | github.com/jrs407/PuppyClicker

- Clicker game built with React 19, TypeScript, Node.js, Express 5 and MySQL 8, with a 10-endpoint REST API, a 7-table relational schema and Docker Compose deployment.
- Server-side game logic with JWT authentication, bcrypt and purchases in transactions with row-level locking (SELECT FOR UPDATE) to prevent double spending.

### Burnout Predictor
May 2026

Machine learning | Python and Spark

- Python model that predicts burnout risk from a set of input criteria, deployed across a Spark node cluster to speed up data processing.

## EDUCATION

### MSc in Computer Engineering
Oct 2025 - Present

University of Almería

Big Data specialisation covering large-scale databases, distributed processing and data pipelines. Fully online and compatible with full-time employment.

### BSc in Computer Engineering
Sep 2021 - Feb 2026

University of Almería

Double specialisation in Software Engineering and Information Systems. Coursework in information security (ISO/IEC 27001), quality management (ISO 9001), risk analysis and GDPR, including a consulting project for a real company.

## LANGUAGES

**Spanish** Native | **English** B2, Cambridge B2 First (FCE)

## STRENGTHS

Teamwork | Analytical thinking | Time management | Continuous learning

## AVAILABILITY

Immediate. On-site, hybrid or remote, and open to relocating anywhere in Spain.